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#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node *next;
};
struct Node *top = NULL;
struct Node *front = NULL;
struct Node *rear = NULL;
struct Node* createNode(int value) {
    struct Node *newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = value;
    newNode->next = NULL;
    return newNode;
}
void push(int value) {
    struct Node *newNode = createNode(value);
    newNode->next = top;
    top = newNode;
    printf("%d pushed to stack.\n", value);
}
void pop() {
    if (top == NULL) {
        printf("Stack is empty. Nothing to pop.\n");
        return;
    }
    struct Node *temp = top;
    printf("%d popped from stack.\n", top->data);
    top = top->next;
    free(temp);
}
void displayStack() {
    struct Node *temp = top;
    if (temp == NULL) {
        printf("Stack is empty.\n");
        return;
    }
    printf("Stack (Top to Bottom): ");
    while (temp != NULL) {
        printf("%d ", temp->data);
        temp = temp->next;
    }
    printf("\n");
}

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void enqueue(int value) {
    struct Node *newNode = createNode(value);
    if (rear == NULL) {
        front = rear = newNode;
    } else {
        rear->next = newNode;
        rear = newNode;
    }
    printf("%d enqueued to the queue.\n", value);
}

void dequeue() {
    if (front == NULL) {
        printf("Queue is empty. Nothing to dequeue.\n");
        return;
    }
    struct Node *temp = front;
    printf("%d dequeued from the queue.\n", front->data);
    front = front->next;
    if (front == NULL)
        rear = NULL;
    free(temp);
}

void displayQueue() {
    struct Node *temp = front;
    if (temp == NULL) {
        printf("Queue is empty.\n");
        return;
    }
    printf("Queue (Front to Rear): ");
    while (temp != NULL) {
        printf("%d ", temp->data);
        temp = temp->next;
    }
    printf("\n");
}

int main() {
    int choice, value, ch;
    while (1) {
        printf("\n--- Singly Linked List Simulation ---\n");
        printf("1. Stack Operations\n");
        printf("2. Queue Operations\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
    }
}

```



```

        scanf("%d", &value);
        enqueue(value);
        break;
    case 2:
        dequeue();
        break;
    case 3:
        displayQueue();
        break;
    case 4:
        goto main_menu;
    default:
        printf("Invalid choice.\n");
    }
}
break;

case 3:
    printf("Exiting program.\n");
    exit(0);

default:
    printf("Invalid choice. Try again.\n");
}

main_menu:;
}
return 0;
}

```

--- Singly Linked List Simulation ---

1. Stack Operations
2. Queue Operations
3. Exit

Enter your choice: 1

--- Stack Menu ---

1. Push
2. Pop
3. Display Stack
4. Back to Main Menu

Enter your choice: 1

Enter value to push: 10

10 pushed to stack.

--- Stack Menu ---

1. Push
2. Pop
3. Display Stack
4. Back to Main Menu

Enter your choice: 1

Enter value to push: 20

20 pushed to stack.

--- Stack Menu ---

1. Push
2. Pop
3. Display Stack
4. Back to Main Menu

Enter your choice: 2

20 popped from stack.

--- Stack Menu ---

1. Push
2. Pop
3. Display Stack
4. Back to Main Menu

Enter your choice: 3

Stack (Top to Bottom): 10

--- Stack Menu ---

1. Push
2. Pop
3. Display Stack
4. Back to Main Menu

Enter your choice: 4

--- Singly Linked List Simulation ---

1. Stack Operations
2. Queue Operations
3. Exit

Enter your choice: 2

--- Queue Menu ---

1. Enqueue
2. Dequeue
3. Display Queue
4. Back to Main Menu

Enter your choice: 1

Enter value to enqueue: 10

10 enqueued to the queue.

--- Queue Menu ---

1. Enqueue
2. Dequeue
3. Display Queue
4. Back to Main Menu

Enter your choice: 1

Enter value to enqueue: 20

20 enqueued to the queue.

--- Queue Menu ---

1. Enqueue
2. Dequeue
3. Display Queue
4. Back to Main Menu

Enter your choice: 1

Enter value to enqueue: 30

30 enqueued to the queue.

--- Queue Menu ---

1. Enqueue
2. Dequeue
3. Display Queue
4. Back to Main Menu

Enter your choice: 2

10 dequeued from the queue.

--- Queue Menu ---

1. Enqueue
2. Dequeue
3. Display Queue
4. Back to Main Menu

Enter your choice: 3

Queue (Front to Rear): 20 30

--- Queue Menu ---

1. Enqueue
2. Dequeue
3. Display Queue
4. Back to Main Menu

Enter your choice: 4

--- Singly Linked List Simulation ---

1. Stack Operations
2. Queue Operations
3. Exit

Enter your choice: 3

Exiting program.

Process returned 0 (0x0) execution time : 36.733 s

Press any key to continue.

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